All of Statistics

Personal Notes

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Contents

1 Probability

Probability is the mathematical language of uncertainty. It begins with a description of every outcome an experiment could produce, then assigns probabilities to collections of those outcomes. Statistical inference later reverses this direction: observed outcomes are used to learn about the process that generated them.

1.1 Sample Spaces and Events

Definition 1.1 (Sample space)**.** The *sample space*, denoted by Ω, is the set of all possible outcomes of an experiment. A particular outcome is written $\omega \in \Omega$. An *event* is a subset $A \subseteq \Omega$.

For one coin toss, $\Omega = \{H, T\}$. For two tosses,

$$\Omega = \{HH, HT, TH, TT\},$$

and the event that the first toss is heads is $A = \{HH, HT\}$. An infinite experiment may have an infinite-dimensional sample space. If a coin is tossed forever, for example,

$$\Omega = \{(\omega_1, \omega_2, \ldots) : \omega_i \in \{H, T\}\}.$$

The event that the first head appears on toss three contains every sequence whose first three entries are T, T, H.

Events obey the usual operations on sets. The complement A^c is the event that A does not occur; $A \cup B$ means that at least one of A and B occurs; and $AB = A \cap B$ means that both occur. The events $A_1, A_2, \ldots$ are *disjoint* or *mutually exclusive* when $A_i A_j = \varnothing$ for $i \neq j$. They form a *partition* of Ω when they are disjoint and their union is all of Ω.

Note: *An outcome is one complete result of the experiment. An event is a claim about that result and may therefore contain many outcomes.*

De Morgan's laws translate logical negation into set notation:

$$\left(\bigcup_i A_i\right)^c = \bigcap_i A_i^c, \qquad \left(\bigcap_i A_i\right)^c = \bigcup_i A_i^c.$$

1.2 Probability Measures

Definition 1.2 (Probability measure)**.** A *probability measure* is a function $\mathbb{P}$ that assigns a real number $\mathbb{P}(A)$ to every event A and satisfies the following axioms:

1. $\mathbb{P}(A) \geq 0$ for every event A;
2. $\mathbb{P}(\Omega) = 1$; and
3. if $A_1, A_2, \ldots$ are disjoint, then

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

The triple $(\Omega, \mathcal{A}, \mathbb{P})$, where $\mathcal{A}$ is the collection of allowed events, is called a *probability space*.

The axioms support two common interpretations. Under the frequency interpretation, $\mathbb{P}(A)$ is the limiting fraction of repeated experiments in which A occurs. Under the degree-of-belief interpretation, it quantifies uncertainty about whether A will occur. The calculations are the same under either interpretation.

Proposition 1.3 (Basic probability rules)**.** *For any events A and B,*

$$\begin{aligned}
\mathbb{P}(\varnothing) &= 0, \\
\mathbb{P}(A^c) &= 1 - \mathbb{P}(A), \\
A \subseteq B &\implies \mathbb{P}(A) \leq \mathbb{P}(B), \\
\mathbb{P}(A \cup B) &= \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(AB).
\end{aligned}$$

Proof. Because Ω and $\varnothing$ are disjoint, $1 = \mathbb{P}(\Omega) = \mathbb{P}(\Omega) + \mathbb{P}(\varnothing)$, so $\mathbb{P}(\varnothing) = 0$. The disjoint union $\Omega = A \cup A^c$ gives the complement rule. If $A \subseteq B$, then $B = A \cup (B \setminus A)$ is a disjoint union, and nonnegativity gives $\mathbb{P}(B) \geq \mathbb{P}(A)$. Finally, decompose $A \cup B$ into the three disjoint events $A \setminus B$, $B \setminus A$, and AB; adding their probabilities counts the intersection once rather than twice. □

Example: If two fair coin tosses are independent and H_i is the event of heads on toss i, then

$$\mathbb{P}(H_1 \cup H_2) = \frac{1}{2} + \frac{1}{2} - \frac{1}{4} = \frac{3}{4}.$$

Probability is continuous along nested sequences of events.

Theorem 1.4 (Continuity of probability). *If $A_1 \subseteq A_2 \subseteq \cdots$, then*

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} \mathbb{P}(A_n).$$

If $A_1 \supseteq A_2 \supseteq \cdots$, then

$$\mathbb{P}\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} \mathbb{P}(A_n).$$

Proof. For an increasing sequence, define $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$ for $n \geq 2$. The B_n are disjoint, $A_n = \bigcup_{i=1}^{n} B_i$, and $\bigcup_n A_n = \bigcup_n B_n$. Countable additivity therefore yields

$$\mathbb{P}\left(\bigcup_{n} A_n\right) = \sum_{i=1}^{\infty} \mathbb{P}(B_i) = \lim_{n\to\infty} \sum_{i=1}^{n} \mathbb{P}(B_i) = \lim_{n\to\infty} \mathbb{P}(A_n).$$

The decreasing result follows by applying the increasing result to $A_1^c \subseteq A_2^c \subseteq \cdots$ and using the complement rule. □

1.3 Finite Sample Spaces

When Ω is finite, a probability measure is determined by the probability of each individual outcome:

$$\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}(\{\omega\}).$$

If all outcomes are equally likely, then each has probability $1/|\Omega|$, and hence

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|}.$$

Probability calculations in this setting become counting problems.

For n distinct objects, there are $n!$ possible orderings. The number of ways to choose an unordered group of k objects is

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Thus, in n fair coin tosses, the probability of exactly k heads is

$$\frac{\binom{n}{k}}{2^n}.$$

1.4 Independent Events

Definition 1.5 (Independence)**.** Two events A and B are *independent*, written $A \perp\!\!\!\perp B$, when

$$\mathbb{P}(AB) = \mathbb{P}(A)\mathbb{P}(B).$$

A collection $\{A_i : i \in I\}$ is independent when

$$\mathbb{P}\left(\bigcap_{i \in J} A_i\right) = \prod_{i \in J} \mathbb{P}(A_i)$$

for every finite subset $J \subseteq I$.

Independence may be a modeling assumption, as with separate coin tosses, or a fact established from the probability measure. For a fair die, the events $A = \{2, 4, 6\}$ and $B = \{1, 2, 3, 4\}$ are independent because

$$\mathbb{P}(AB) = \frac{2}{6} = \frac{1}{2} \cdot \frac{2}{3} = \mathbb{P}(A)\mathbb{P}(B).$$

Note: *Disjointness and independence describe different ideas. Disjoint positive probability events cannot be independent: disjointness forces* $\mathbb{P}(AB) = 0$*, while independence would require* $\mathbb{P}(AB) = \mathbb{P}(A)\mathbb{P}(B) > 0$.

Example: For ten independent fair tosses, the probability of at least one head is most easily found through the complement:

$$1 - \mathbb{P}(\text{all tails}) = 1 - \left(\frac{1}{2}\right)^{10}.$$

Pairwise independence alone is weaker than independence of a collection. The full definition requires the product rule for every finite subcollection, not only for each pair.

1.5 Conditional Probability

Definition 1.6 (Conditional probability)**.** If $\mathbb{P}(B) > 0$, the *conditional probability* of A given B is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(AB)}{\mathbb{P}(B)}.$$

For fixed B, the map $\mathbb{P}(\cdot \mid B)$ is itself a probability measure. Conditioning restricts attention to the outcomes in B, then asks what fraction also lie in A. Rearranging the definition gives the multiplication rule

$$\mathbb{P}(AB) = \mathbb{P}(A \mid B)\mathbb{P}(B) = \mathbb{P}(B \mid A)\mathbb{P}(A).$$

Consequently, when $\mathbb{P}(B) > 0$, independence is equivalent to $\mathbb{P}(A \mid B) = \mathbb{P}(A)$: learning that B occurred does not change the probability of A.

Note: *In general, $\mathbb{P}(A \mid B) \neq \mathbb{P}(B \mid A)$. Confusing these two directions is the prosecutor's fallacy. A test may be likely to return positive when a disease is present even though the disease remains unlikely after a positive result, especially when the disease is rare.*

Example: Suppose a disease has probability .01, a test is positive with probability .9 for a sick person, and it is positive with probability .1 for a healthy person. Then

$$\mathbb{P}(D \mid +) = \frac{(.9)(.01)}{(.9)(.01) + (.1)(.99)} = \frac{.009}{.108} \approx .083.$$

The base rate of the disease is essential to the calculation.

Repeated use of the multiplication rule produces the chain rule. For example,

$$\mathbb{P}(ABC) = \mathbb{P}(A \mid BC)\mathbb{P}(B \mid C)\mathbb{P}(C).$$

1.6 The Law of Total Probability and Bayes' Theorem

Theorem 1.7 (Law of total probability)**.** *Let $A_1, \ldots, A_k$ be a partition of Ω with $\mathbb{P}(A_i) > 0$. For any event B,*

$$\mathbb{P}(B) = \sum_{i=1}^{k} \mathbb{P}(B \mid A_i)\mathbb{P}(A_i).$$

Proof. The events $BA_1, \ldots, BA_k$ are disjoint and their union is B. Therefore

$$\mathbb{P}(B) = \sum_{i=1}^{k} \mathbb{P}(BA_i) = \sum_{i=1}^{k} \mathbb{P}(B \mid A_i)\mathbb{P}(A_i).$$

□

Theorem 1.8 (Bayes' theorem)**.** *Let $A_1, \ldots, A_k$ be a partition of Ω with $\mathbb{P}(A_i) > 0$. If $\mathbb{P}(B) > 0$, then*

$$\mathbb{P}(A_i \mid B) = \frac{\mathbb{P}(B \mid A_i)\mathbb{P}(A_i)}{\sum_{j=1}^{k} \mathbb{P}(B \mid A_j)\mathbb{P}(A_j)}.$$

Proof. By the multiplication rule and the law of total probability,

$$\mathbb{P}(A_i \mid B) = \frac{\mathbb{P}(A_iB)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B \mid A_i)\mathbb{P}(A_i)}{\sum_{j=1}^{k} \mathbb{P}(B \mid A_j)\mathbb{P}(A_j)}.$$ □

The value $\mathbb{P}(A_i)$ is the *prior probability* of A_i, and $\mathbb{P}(A_i \mid B)$ is its *posterior probability* after observing B. Bayes' theorem converts the forward probability of evidence under a hypothesis into the reverse probability of the hypothesis given the evidence.

Example: Suppose email is spam, low priority, or high priority with prior probabilities .7, .2, and .1. The word "free" appears with conditional probabilities .9, .01, and .01 in those categories. For an email containing that word,

$$\mathbb{P}(\text{spam} \mid \text{"free"}) = \frac{(.9)(.7)}{(.9)(.7) + (.01)(.2) + (.01)(.1)} \approx .995.$$

1.7 Measurable Events

On an uncountable sample space, assigning probabilities consistently to every subset is generally impossible. Probability theory therefore specifies a collection $\mathcal{A}$ of measurable events.

Definition 1.9 (σ-algebra)**.** A collection $\mathcal{A}$ of subsets of Ω is a *σ-algebra* when:

1. $\varnothing \in \mathcal{A}$;
2. $A \in \mathcal{A}$ implies $A^c \in \mathcal{A}$; and
3. $A_1, A_2, \ldots \in \mathcal{A}$ implies $\bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$.

These conditions guarantee closure under the event operations used by probability. On $\mathbb{R}$, the standard choice is the *Borel σ-algebra*, the smallest σ-algebra containing every open set.

Source. These notes synthesize Chapter 1, "Probability," of Larry Wasserman's *All of Statistics: A Concise Course in Statistical Inference* (Springer, 2004).